

Activity: exponential dispersion models

Exponential dispersion models

We say that a distribution belongs to the *exponential dispersion model* family if the distribution's probability function can be written in the form

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

where θ is called the *canonical parameter*, $\kappa(\theta)$ is the *cumulant function*, $\phi > 0$ is the *dispersion parameter*, and $a(y, \phi)$ is a function – which does not depend on θ – that ensures $f(y; \theta, \phi)$ integrates or sums to 1.

Normal distribution

The $N(\mu, \sigma^2)$ distribution has pdf

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (y - \mu)^2 \right\}$$

Show that the $N(\mu, \sigma^2)$ distribution belongs to the EDM family.